Explainable Multi-Class Thyroid-Disease Classification from Laboratory Panels: A Detailed Study of Tree-Ensemble Models, Their Training, and Per-Prediction SHAP Interpretation

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***Abstract*—Thyroid dysfunction is among the most common endocrine disorders, and its diagnosis rests almost entirely on a small panel of laboratory measurements governed by a well-understood feedback loop. This makes it an ideal testbed for machine-learning models that must be both accurate and *explainable*. We present a detailed, reproducible framework that classifies patients into seven clinically coherent thyroid categories from the Garavan Institute archive (n=9,172), a large, severely imbalanced (73.8% majority) dataset with informative missing laboratory values. We study five tree-ensemble models in depth—Random Forest, XGBoost, a soft-voting hybrid of the two, LightGBM, and a stacking ensemble—explaining the learning principle, the optimisation objective, and the exact training procedure of each. We describe a leakage-safe preprocessing and cross-validation methodology, an explicit treatment of class imbalance, and a statistical protocol for comparing models. The hybrid attains a hold-out accuracy of 0.945 and macro-F1 of 0.863, with a paired t-test confirming a statistically significant improvement over the Random-Forest baseline (p=0.014). We then use exact TreeSHAP to produce both global feature attributions and *detailed per-prediction* explanations that recover textbook thyroid physiology (e.g. suppressed TSH with elevated free-thyroxine index driving a hyperthyroid prediction). The result is a transparent pipeline whose every decision can be audited against clinical reasoning.**

***Index Terms*—Thyroid disease, Random Forest, XGBoost, LightGBM, stacking, soft voting, SHAP, explainable AI, class imbalance, gradient boosting.**

I. Introduction

The thyroid gland regulates basal metabolic rate through the hormones thyroxine (T4) and triiodothyronine (T3), whose secretion is controlled by pituitary thyroid-stimulating hormone (TSH) under negative feedback. Because a handful of serum measurements (TSH, T3, total T4, thyroxine-binding capacity, and the free thyroxine index) capture this axis almost completely, thyroid status is a problem where a machine-learning model's reasoning can be checked directly against physiology. This is rarely true in clinical ML, and it is the central motivation for this study.

We address a seven-class formulation that distinguishes the normal state from hypothyroidism, hyperthyroidism, binding-protein anomalies, non-thyroidal illness, ongoing replacement therapy, and discordant assay results. The dataset is large (9,172 patients), heavily imbalanced, and contains laboratory values that are *missing not at random*—a panel is ordered when a clinician suspects a problem, so the absence of a measurement is itself informative. These properties make the problem a realistic stress test rather than a sanitised benchmark.

This paper emphasises *depth* on two fronts that are often compressed in applied studies: (i) a careful account of every model and exactly how it is trained, and (ii) per-prediction explanations. Our contributions are a detailed comparative treatment of five tree ensembles and their training methodology; a leakage-safe, imbalance-aware evaluation protocol; a statistically grounded model comparison; and exact SHAP explanations at both the global and the individual- patient level.

II. Clinical Background

A. The Hypothalamic–Pituitary–Thyroid Axis

The hypothalamus secretes thyrotropin-releasing hormone, prompting the pituitary to release TSH, which stimulates the thyroid to produce T4 and T3. Circulating thyroid hormone then suppresses further TSH release—a thermostat. When the *gland* fails, hormone falls and TSH rises to compensate; when the gland is overactive, hormone rises and TSH is suppressed. TSH and thyroid hormone therefore move in *opposite* directions in primary disease, and because TSH responds logarithmically it is the single most sensitive screening test. A model trained on these features should, and does, lean heavily on TSH.

B. The Seven Target Classes

We group the archive's raw diagnosis codes into: *negative* (no thyroid condition), *hypothyroid* (under-active; high TSH, low free T4), *hyperthyroid* (over-active; suppressed TSH, high T4/T3), *binding_protein* (altered thyroxine-binding globulin, e.g. from pregnancy or estrogen, shifting *total* but not free hormone), *nonthyroidal_illness* (the "sick-euthyroid" pattern, typically low T3 in seriously ill patients), *replacement_therapy* (patients on exogenous thyroxine), and *discordant_results* (assays that disagree). Several of these classes are physiologically subtle, which is why a clean test of an explainable model is valuable.

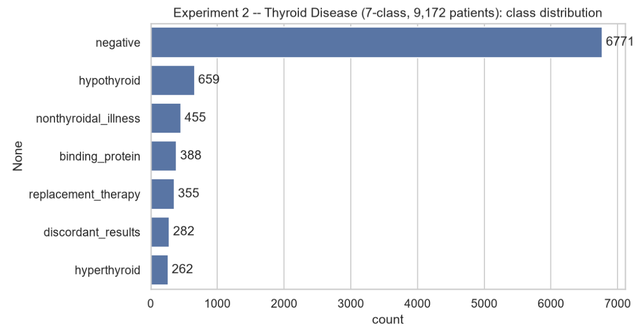

Fig. 1. Class distribution of the seven thyroid groups. The problem is severely imbalanced, motivating macro-averaged metrics and imbalance-aware training.

III. Dataset and Preprocessing

A. Data

The Garavan `thyroid0387` archive contains 9,172 records with 29 attributes and a diagnosis string. Six attributes are continuous laboratory values (age, TSH, T3, total T4, T4U, FTI); fourteen are binary clinical-history flags (e.g. *on_thyroxine*, *pregnant*, *lithium*, *thyroid_surgery*); two are nominal (sex, referral source). The class distribution (Fig. 1) is dominated by the negative class (73.8%), with the rarest disease class (hyperthyroid) at 2.9%.

B. Cleaning, Encoding and Missingness

We map the diagnosis codes to the seven groups above, taking the more-likely label where the archive records an ambiguous "X|Y" code, and folding the small antithyroid-treatment codes into the hyperthyroid group (they denote hyperthyroid management). Binary flags are mapped $\{f, t\} \rightarrow \{0, 1\}$; a non-physiological age >120 is treated as missing; the near-empty TBG measurement and the redundant "measured" indicator columns are dropped.

Crucially, several laboratory values are missing for a large fraction of patients (T3 in 28%, TSH in 9%). Rather than discard these rows, we *median-impute* each numeric feature and append a binary *missing-indicator* column so the model can learn from the fact that a test was not ordered. Nominal features are most-frequent imputed and one-hot encoded. Because tree ensembles are invariant to monotone feature scaling, no standardisation is applied. The complete transform is encapsulated in a `ColumnTransformer` that is fit only on training data (Section V-A).

IV. Models

All five models are tree-based, a deliberate choice: trees handle mixed feature types and missing-indicator flags natively, require no scaling, capture the threshold logic of clinical decision-making ("TSH>6 *and* FTI<60"), and admit *exact* SHAP explanations. We now describe each model, its learning principle, and its training.

A. Foundation: the Decision Tree

A decision tree partitions feature space by recursively choosing the feature/threshold that most reduces node impurity, measured for a node with class proportions p_c by the Gini index $G = 1 - \sum_c p_c^2$ or entropy $H = -\sum_c p_c \log p_c$. A leaf predicts the empirical class distribution of its training samples. A single deep tree has low bias but high variance: it memorises the training set. The two ensemble families below tame this variance and bias in complementary ways, which is exactly why combining them is attractive.

B. Random Forest (Bagging)

A Random Forest averages T decorrelated trees,

$$f_{\mathrm{RF}}(x) = \frac{1}{T} \sum_{t=1}^{T} h_t(x), \tag{1}$$

where each h_t is grown on a *bootstrap* resample of the training data (rows drawn with replacement) and, at every split, considers only a random subset of features (typically $\sqrt{d}$). These two randomisations decorrelate the trees. If individual trees have variance σ^2 and pairwise correlation ρ, the average has variance

$$\rho\,\sigma^2 + \frac{1-\rho}{T}\,\sigma^2, \tag{2}$$

so lowering ρ (via the randomisation) and increasing T both reduce variance without raising bias—the essence of *bagging*. For multi-class problems each tree outputs a class distribution and the forest averages them, giving calibrated `predict_proba` outputs that the hybrid later consumes. We use T=300 fully grown trees. Imbalance is handled with `class_weight = balanced_subsample`, which reweights classes within each bootstrap (Section V-C).

C. XGBoost (Gradient Boosting)

Where bagging reduces variance, *boosting* reduces bias by building trees *sequentially*, each correcting its predecessors. XGBoost fits an additive model $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$ by minimising, at step t, the regularised objective

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} \ell\Big(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\Big) + \Omega(f_t), \tag{3}$$

with complexity penalty $\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_j w_j^2$ over a tree's T leaves and leaf weights w_j. Expanding the loss to second order with gradients $g_i = \partial_{\hat{y}}\ell$ and Hessians $h_i = \partial^2_{\hat{y}}\ell$ gives the per-leaf optimum and the resulting structure score

$$w_j^\star = -\frac{G_j}{H_j + \lambda}, \qquad \tilde{\mathcal{L}} = -\frac{1}{2}\sum_j \frac{G_j^2}{H_j + \lambda} + \gamma T, \quad (4)$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ aggregate over the samples in leaf j. A candidate split is scored by the gain

$$\text{Gain} = \tfrac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma, \quad (5)$$

and accepted only if positive—this is how λ and γ prevent overfitting at the level of individual splits. We use the multi-class softmax objective (`multi:softprob`), 400 trees of depth 6, a learning rate (shrinkage) of 0.1 that scales each tree's contribution, row and column subsampling of 0.9 (borrowing bagging's decorrelation), λ=1, and the fast histogram split-finder. Because XGBoost has no `class_weight`, we pass *balanced per-sample weights* (Section V-C).

D. The Hybrid: Soft Voting

The hybrid combines the variance-reducing forest and the bias-reducing booster by averaging their posterior probabilities and taking the arg-max,

$$\hat{y} = \arg\max_{c \in \{1,\dots,K\}} \frac{1}{M}\sum_{m=1}^{M} P_m(c \mid x), \qquad M = 2. \quad (6)$$

This is *soft* voting: it uses each model's confidence, so a decisive model sways the average more than a hesitant one. The two learners' errors are partly uncorrelated, so the average cancels some of each. Both base learners sit behind a single shared preprocessing transform, guaranteeing identical inputs.

E. LightGBM

LightGBM is a second gradient booster with two efficiency innovations. *Leaf-wise* (best-first) growth splits the leaf with the largest loss reduction rather than expanding a whole level, producing deeper, more accurate trees for the same leaf budget. *Histogram* binning buckets continuous features to accelerate split finding. It further uses gradient-based one-side sampling (keeping large-gradient samples while sub-sampling small-gradient ones) and exclusive-feature bundling (merging sparse, mutually-exclusive features). We use 400 trees, 63 leaves, and a learning rate of 0.05, with `class_weight = balanced`.

F. Stacking

Soft voting combines models with *fixed equal* weights. Stacking instead *learns* the combination. Level-0 base learners (Random Forest, XGBoost, LightGBM) are trained, and their *out-of-fold* predicted probabilities— obtained by internal 5-fold cross-validation so the meta-features are leakage-free— become the inputs to a level-1 meta-learner, here a multinomial logistic regression with balanced class weights. The meta-learner can discover, for example, that XGBoost is more trustworthy for one class while the forest is better for another, an adaptivity that fixed-weight voting cannot express. The cost is more computation and a greater risk of overfitting, which we limit by keeping the meta-learner linear.

V. Training Methodology

A. Leakage-Safe Pipeline

Every model is a scikit-learn `Pipeline` whose first stage is the preprocessing `ColumnTransformer` and whose second stage is the classifier. When the pipeline is fit, imputation medians and one-hot vocabularies are learned from the *training partition only*; at validation/test time these learned transforms are merely applied. Because cross-validation clones and refits the entire pipeline on each fold, no statistic computed on a held-out fold can leak into training—a frequent and silent source of over-optimistic results that this design eliminates by construction (Fig. 2).

B. Stratified Splitting and Cross-Validation

We hold out 20% of patients as a test set with *stratified* sampling, which preserves each class's proportion; on a problem where the rarest class is 2.9% of data, a naive split could leave that class with too few test cases to estimate recall. On the remaining 80% we run stratified five-fold cross-validation, training each model five times and reporting the mean and standard deviation of each metric, with the fold standard deviation

$$\sigma = \sqrt{\tfrac{1}{4}\sum_{j=1}^{5}(S_j - \mu)^2}. \quad (7)$$

We retain the *individual* fold scores because the significance test (Section VI) compares two models fold-by-fold on identical splits.

C. Class-Imbalance Handling

With a 73.8% majority, a model can score high accuracy while ignoring every disease. We counter this in two equivalent ways. For Random Forest, LightGBM and the stacking meta-learner we set balanced class weights

$$w_c = \frac{N}{K\, n_c}, \quad (8)$$

which up-weights a class inversely to its frequency n_c. XGBoost, lacking a class-weight argument, receives the same correction as *per-sample* weights $w_i = w_{y_i}$ at fit time. (Synthetic minority over-sampling, applied inside the training fold, is supported as an alternative.) Coupled with macro-averaged metrics, this ensures rare diseases materially affect both training and scoring.

D. Hyperparameter Configuration

Table I lists the configuration of every model. These are strong, lightly-tuned defaults chosen for a fair comparison rather than exhaustively optimised; all share a fixed random seed (42) for reproducibility.

TABLE I
MODEL HYPERPARAMETERS.

Model	Key settings
Random Forest	300 trees, unlimited depth, $\sqrt{d}$ features/split, `balanced_subsample` weights
XGBoost	400 trees, depth 6, lr 0.1, subsample 0.9, colsample 0.9, λ=1, `multi:softprob`, hist, balanced sample weights
LightGBM	400 trees, 63 leaves, lr 0.05, subsample 0.9, colsample 0.9, `balanced` weights
Hybrid	soft voting of the above RF and XGBoost (M=2, equal weights)
Stacking	base: RF, XGBoost, LightGBM; meta: multinomial logistic regression (5-fold out-of-fold features)

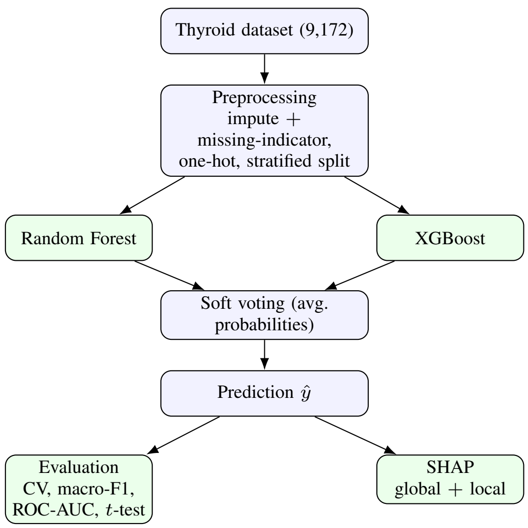

Fig. 2. Leakage-safe training and evaluation pipeline. Preprocessing is fit inside each cross-validation fold.

E. Environment and Reproducibility

Experiments use Python with scikit-learn, XGBoost, LightGBM and the SHAP library. All randomness (splits, bootstraps, column sampling, fold shuffling) is seeded; the data download, preprocessing, training, evaluation and figure generation run from a single scripted pipeline so the entire study is reproducible end-to-end.

VI. EVALUATION PROTOCOL

On a heavily imbalanced problem, plain accuracy is misleading—predicting the majority class for everyone already scores 0.738. We therefore foreground *macro*-averaged F1, which computes F1 per class and averages classes equally:

$$\mathrm{F1}_i = \frac{2\,\mathrm{Prec}_i\,\mathrm{Rec}_i}{\mathrm{Prec}_i + \mathrm{Rec}_i}, \qquad \text{Macro-F1} = \frac{1}{K}\sum_{i=1}^{K}\mathrm{F1}_i. \quad (9)$$

We also report balanced accuracy (mean per-class recall) and one-vs-rest ROC–AUC, which uses predicted probabilities and so measures class separation independent of any threshold. To test whether the hybrid genuinely beats the Random-Forest baseline we apply a *paired* t-test to the fold-wise macro-F1 scores,

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad n = 5, \quad (10)$$

where d_i is the per-fold difference. Pairing on identical folds removes fold-to-fold variance and makes the test sensitive to the model difference itself.

TABLE II
STRATIFIED FIVE-FOLD CROSS-VALIDATION, MEAN $\pm$ STD.

Model	Accuracy	Macro-F1	ROC-AUC
Random Forest	0.941 ± 0.008	0.850 ± 0.024	0.992
XGBoost	0.950 ± 0.006	0.872 ± 0.020	0.995
Hybrid (RF+XGB)	$\mathbf{0.952 \pm 0.005}$	$\mathbf{0.875 \pm 0.018}$	0.995
LightGBM	0.951 ± 0.005	0.873 ± 0.018	0.994
Stacking	0.951 ± 0.005	0.874 ± 0.015	0.991

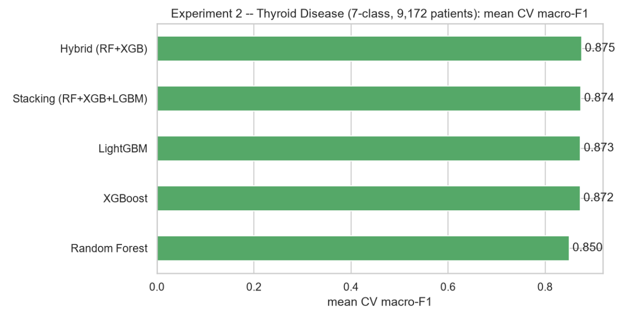

Fig. 3. Mean cross-validated macro-F1 by model.

VII. RESULTS

A. Cross-Validation and Model Comparison

Table II reports five-fold cross-validation for all five models, and Fig. 3 visualises their macro-F1. The hybrid attains the best mean macro-F1 (0.875) with the smallest variance among the classic trio, echoing the expectation that soft voting improves *stability*. The two boosters (XGBoost, LightGBM) and stacking are close behind; the gain over plain XGBoost is small, so most of the lift over the forest comes from boosting rather than from voting *per se*.

B. Statistical Significance

The paired t-test of hybrid versus Random Forest gives $\bar{d}$=+0.0255, t=4.16, p=0.014: the improvement *is* statistically significant at the 0.05 level. The large sample size (9,172 patients, hence ample fold data) supplies the statistical power needed to resolve a genuine-but-modest effect—a contrast with smaller studies where the same kind of improvement is undetectable.

C. Hold-Out Test Performance

On the unseen 20% test set the hybrid attains accuracy 0.945, balanced accuracy 0.850, macro-F1 0.863, weighted-F1 0.944 and ROC–AUC 0.995. The gap between accuracy

TABLE III
PER-CLASS HOLD-OUT PERFORMANCE OF THE HYBRID.

Class	Prec.	Rec.	F1	Support
negative	0.963	0.974	0.969	1355
hypothyroid	0.949	0.985	0.967	132
nonthyroidal illness	0.977	0.923	0.949	91
binding protein	0.829	0.808	0.818	78
replacement therapy	0.903	0.915	0.909	71
discordant results	0.738	0.554	0.633	56
hyperthyroid	0.804	0.789	0.796	52
macro avg	0.880	0.850	0.863	1835

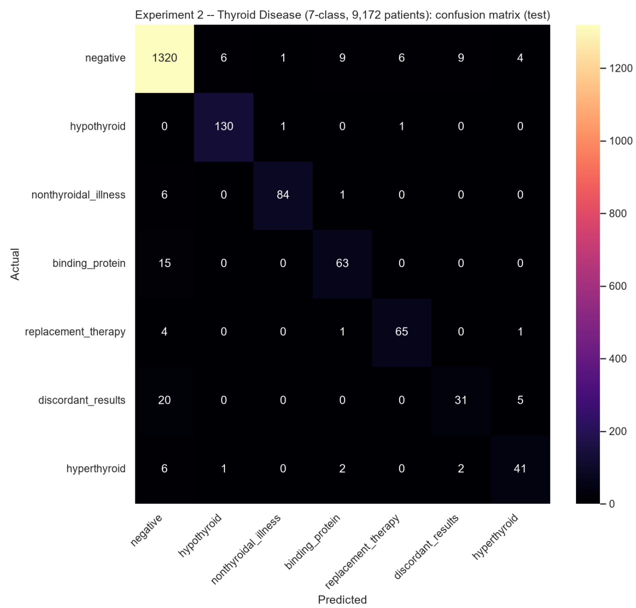

Fig. 4. Confusion matrix of the hybrid on the thyroid hold-out set (counts).

and macro-F1 is the imbalance signature: excellent on large classes, merely good on rare ones. Per-class results (Table III) show near-perfect hypothyroid detection (F1 0.967), the unambiguous high-TSH signal being easy to learn, and the weakest performance on the intrinsically ambiguous discordant-results class (F1 0.633)—a class defined by disagreeing assays where, by construction, no clean signal exists. The confusion matrices (Figs. 4, 5) and ROC curves (Fig. 6) confirm strong, well-separated performance across classes.

VIII. EXPLAINABILITY

A. Global Attribution

For a tree model and target class, SHAP decomposes a prediction additively, $f(x) = \phi_0 + \sum_j \phi_j$, with base value ϕ_0 and signed per-feature contributions ϕ_j computed exactly by TreeSHAP. Averaging $|\phi_j|$ over all patients yields the global importances in Fig. 7: TSH, T3, FTI, total T4, *on_thyroxine* and T4U lead, exactly the thyroid function panel in clinically sensible order. The SHAP *interaction* summary (Fig. 8) decomposes each contribution into a main effect (the matrix

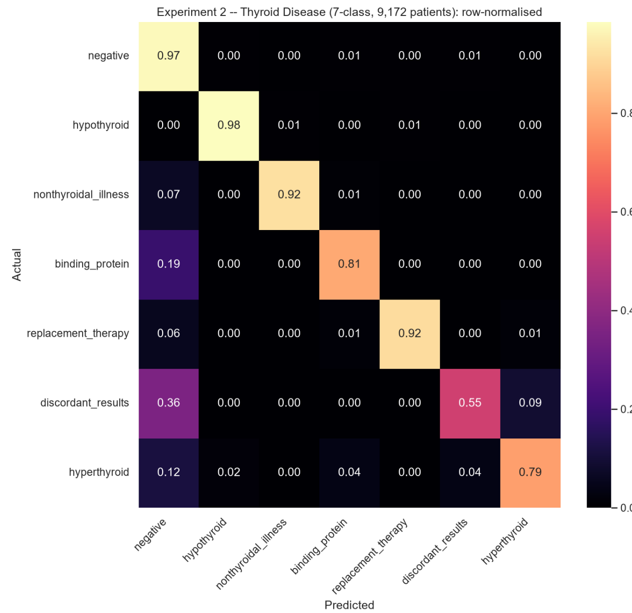

Fig. 5. Row-normalised confusion matrix (per-class recall on the diagonal).

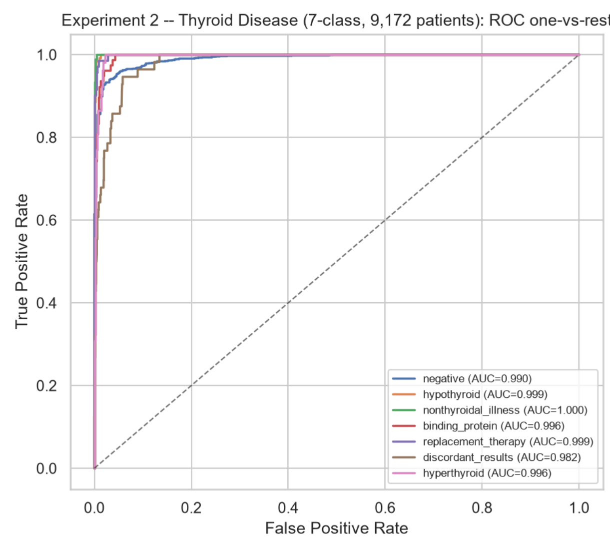

Fig. 6. One-vs-rest ROC curves per class for the hybrid.

diagonal) and pairwise interaction effects (off-diagonal cells): the dominant cells involve TSH, total T4 and FTI, and the TSH×FTI and TSH×TT4 interactions encode the clinical rule that TSH must be interpreted *together* with a thyroid-hormone level rather than alone. The high rank of the T3 and TSH missing-indicators confirms that *which* tests were ordered is itself diagnostic.

B. Per-Prediction Explanations

The additive decomposition lets us narrate individual predictions. Three representative correctly-classified patients:

- **Hyperthyroid** (Fig. 9): a very high free-thyroxine index (FTI=196, +3.21) and a suppressed TSH (0.02, +2.21)

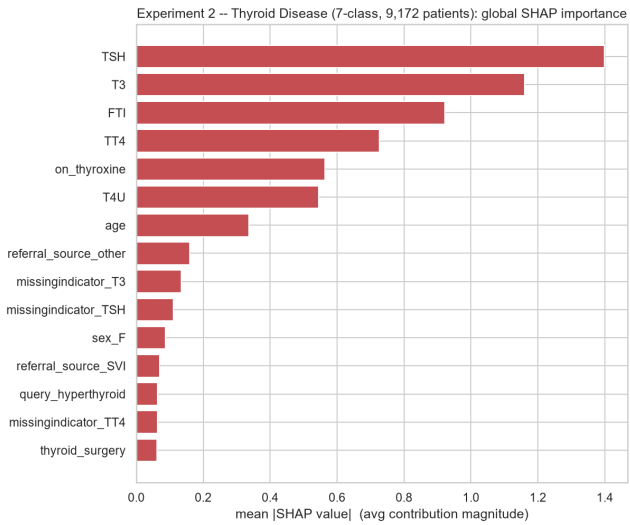

Fig. 7. Global SHAP importance (mean $|\phi_j|$) of the XGBoost component.

dominate— the textbook over-active-gland signature.

- **Hypothyroid** (Fig. 10): an elevated TSH (6.4, $+3.80$) overwhelmingly drives the prediction, as the feedback loop predicts.
- **Replacement therapy** (Fig. 11): the *on_thyroxine* flag ($+3.62$) plus a high FTI (164, $+2.45$) identify a patient taking exogenous hormone—a case where a clinical-history flag, not a single lab, is decisive.

Because the contributions sum to the model's score, each narrative is a complete, auditable account rather than a post-hoc story. Applied to misclassified patients, the same decomposition reveals *why* the model erred (typically a borderline lab value combined with a missing T3), turning silent errors into diagnosable ones.

IX. Discussion

The study shows that a carefully trained tree-ensemble pipeline classifies seven thyroid states with high accuracy and, more importantly, with reasoning that matches physiology at the level of individual patients. The soft-voting hybrid is the best of the classic trio and—unlike in smaller studies—its advantage over the Random-Forest baseline is statistically significant, an effect attributable to sample size rather than to a fundamentally different model. The principal difficulty is the discordant-results class, where the model is appropriately uncertain; this is a feature, not a bug, of an honest evaluation. The treatment of informative missingness via indicator features, surfaced as genuinely predictive by SHAP, is a practical lesson for laboratory-based ML.

X. Limitations

The archive is single-institution and decades old; external multi-centre validation is required before clinical use. The seven-class grouping merges fine-grained diagnoses and folds antithyroid-treatment codes into hyperthyroid. Hyperparameters were only lightly tuned. SHAP attributions are associational, not causal. Rare classes remain under-represented, limiting the reliability of their per-class estimates.

XI. Conclusion

We presented a detailed, reproducible, and explainable framework for seven-class thyroid-disease classification, with an in-depth account of five tree-ensemble models and exactly how they are trained, an imbalance-aware and leakage-safe methodology, a statistically grounded comparison, and exact per-prediction SHAP explanations that recover thyroid physiology. The hybrid achieves accuracy 0.945 and macro-F1 0.863, with a significant improvement over the baseline. Future work includes external validation, cost-sensitive learning for rare classes, SHAP interaction analysis, and probability calibration for actionable confidence.

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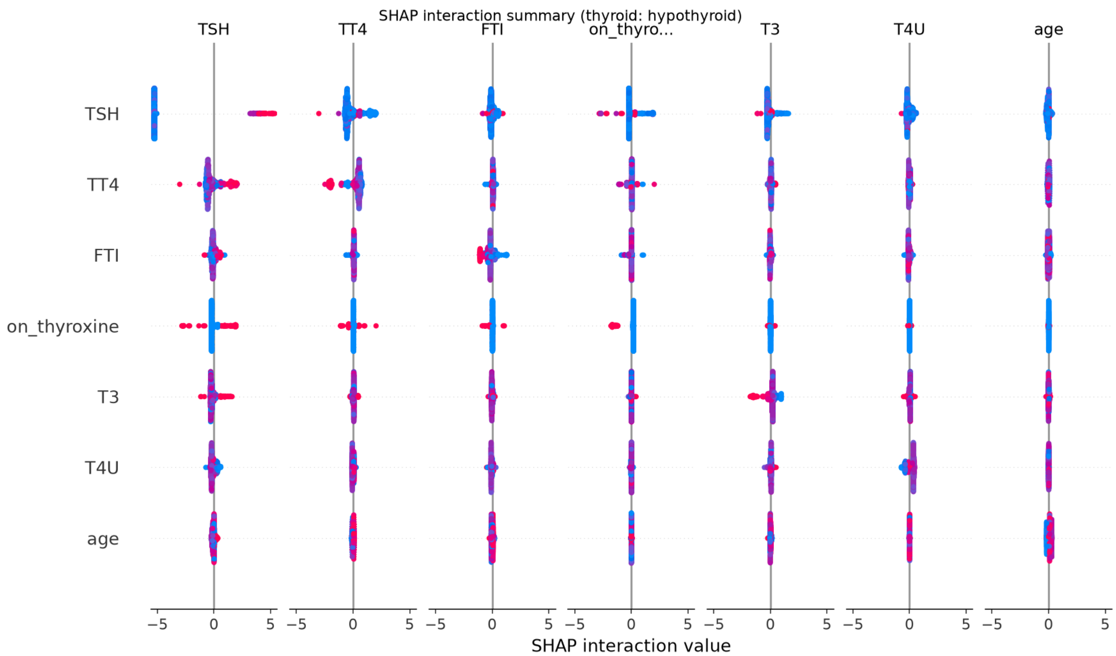

Fig. 8. SHAP interaction-value summary for the hypothyroid output. The top seven features appear on both axes; each cell is a beeswarm of SHAP *interaction* values (x-axis), with colour encoding the feature value. Diagonal cells are a feature's main effect; off-diagonal cells are pairwise interactions.

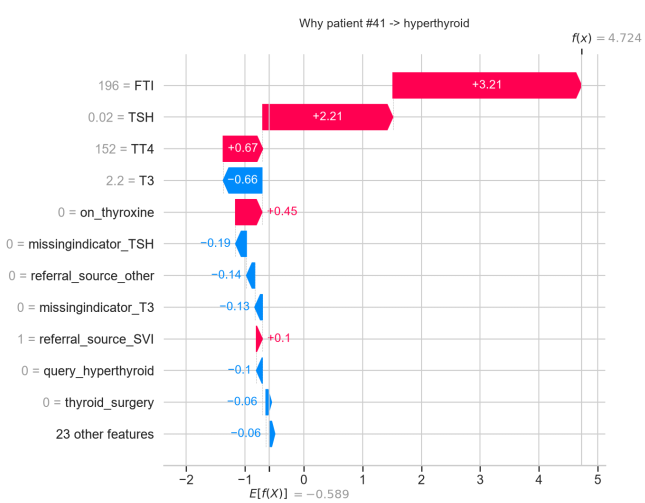

Fig. 9. Per-prediction SHAP waterfall: hyperthyroid. FTI and suppressed TSH dominate.

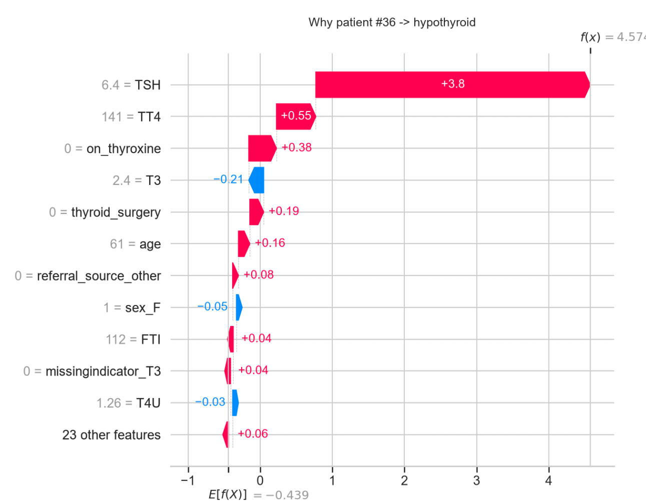

Fig. 10. Per-prediction SHAP waterfall: hypothyroid. Elevated TSH dominates.

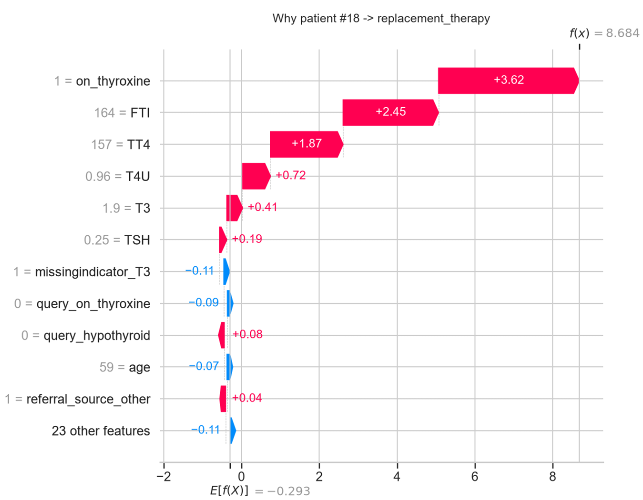

Fig. 11. Per-prediction SHAP waterfall: replacement therapy. The *on_thyroxine* history flag is decisive.